Service Reservation, Labor & Equipment Shifting Management System for Construction

Abstract

The organization currently has a manual paper-based system. In above activities through the doing in manual process, it is very inconsistent, time consuming & lot of paper waste. And there is no proper mechanism manage service reservation details & difficult to maintain the BSR details (Building scheduling rates) for every year & every province as well as difficult to doing hard calculation for estimation.

The proposed system basically focusses on making an easy to way service reservation process & maintain the BSR details. As well as through the proposed system can be calculate the estimation and making house plan for relevant reservation. And it will help the reduce hard calculation of these processes. Through the employee scheduling process can be track the employee's working progress in certain period of time. And as well as managing the equipment transfer information among reservations. There for proposed system lot of minimize their wasting time and drawbacks.

The proposed system was developed using the object-oriented concepts, followed the RESET architecture, MVC architecture & Hibernate technologies. The REST server was created through the Spring frame work. The MVC architecture lot of contribute the maintain the solution and minimize the drawbacks. JAVA is selected for the programming language and it has contributed the follow object-oriented concepts. MYSQL is used for handling the database and IntelliJ IDEA as used for the Integrated Development Environment (IDE). All UML diagrams drawn through the Visual Paradigm. And the Iterative and incremental development model for use as the development methodology.

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List of acronyms

BOQ – Bill of Quantities

BSR – Building Scheduling Rates

CBIS – Computer Base Information System

DAO – Data Access Object

GUI – Graphical User Interface

IT – Information Technology

MVC – Model-View-Controller

QS – Quantity Surveyor

UI – User Interface

Chapter 1 - Introduction

Ceylon Engineering company is the market leader of construction field in Sri Lanka & located at Moratuwa. Most company project were & are in Colombo district & island wide. They are providing the residence planning, designing, consulting, passing counselling approvals, preparation BOQ, handling the QS part & providing quality standard construction service for customers.

1.1 Motivation for the project

The Construction of Ceylon Engineering company currently carry out their activities in base on manually processes. In these processes are very inconsistence, time consuming & lot of paper waste. Lot of companies have followed CBIS for management system in these days. In here they usually follow up manual process & because of that they face lot of problems to their working process.

The problem encountered with their manual process is,

- There is no proper mechanism to manage the service & reserve details.
- There is no proper way to manage customer details & their payments.
- Employee scheduling process is very inconsistent & time consuming.
- Estimation details maintaining process very inconsistent & time consuming.
- Difficult to maintaining BSR details process for every year & every province.
- Lot of paper wastage, wasting time to find some details & sometimes destroy the papers in due to manual process.
- Spending long duration to equipment transfer process.
- Difficult to generate equipment allocation reports, employee allocation reports, previous estimation reports & previous BSR reports.

1.2 Objectives

The main goal of the proposed system is built up the efficient & accurate process for the continue company activities. The CBIS was improved the efficiency of their important activities. Minimize their wasting time & drawbacks. Furthermore, increase the accurately of working process in their objectives.

Other objectives & scopes are listed below,

- Making an easy to way service reservation process.
- Improve efficiency & accurate customer details & payment management process. By it easy to check the customer payment balance.
- Solve the employee scheduling problem & improving efficiency of the employee scheduling process.
- Improving & maintain the estimation details without hard manual calculation.
- Making an easy way to maintain BSR details.
- When using the Database approach, it will be reduced paper cost less than 60% precent of wastage.
- It 50% precent able to saving time for spending equipment transfer process.
- Easy to generate reports without any complexity.

1.3 Scope

Following are the scope of the project,

a) Customer Manage
    - Manage Customer's main details.
    - Manage Customer Payment details.
    - Manage Customer's reservation progress details.

b) Reservation Manage
    - Manage BSR details.
    - Manage Estimation details.
    - Project plan management details.

c) Transfer Manage

- Manage Employee Transfer details.
- Manage Equipment details.
- Manage Equipment Transfer details.

d) Administration

- Employee details management.
- User details management.
- Privilege details management.
- Main Window & Login management.
- Report & Notification management.

1.4 Structure of the dissertation

Chapter 02 - Analysis

This chapter will provide details about the analysis phase of the system development life cycle. A collected good set of requirements including both functional and nonfunctional requirement are listed in this chapter.

Chapter 03 - Design

After requirement elicitation, the design should be done. Details about the design phase, diagram which are drawn by us including use case, class, database, activity are drawn in this chapter.

Chapter 04 - Evaluation

Before the system is delivered into the working environment, it should be tested properly. This chapter will be discussed about testing techniques, test case and more and also this chapter will prove that proper testing was done.

Chapter 05 - Conclusion

This chapter will conclude the entire dissertation report with the evaluation.

Chapter 2 – Analysis

Requirements analysis or requirements engineering is a process used to determine the needs and expectations of a new product. It involves frequent communication with the stakeholders and end-users of the product to define expectations, resolve conflicts, and document all the key requirements. [1]

One of the greatest challenges faced by any organization is to share the vision of the final product with the customers. Hence, a business requirements analysis involves a team effort of all the key stakeholders, software developers, end-users, and customer managers to achieve a shared understanding of what the product should do. [1].

2.1 Review of similar systems

Reviewing of similar system more help to the understand proposed system & how to identify the functionalities should be presented. Following reviewed of similar systems were use to developed for the proposed system.

2.1.1 DANAOS Construction

To identify the similar system, for researched some similar system. The DANAOS construction management system is more similar with this proposed system. The DANOS construction management system is proprietary & web-based software. It allows roles, user & groups creation providing secure & regulated control.

Following figure 2.1 shows the dashboard of DANAOS Construction

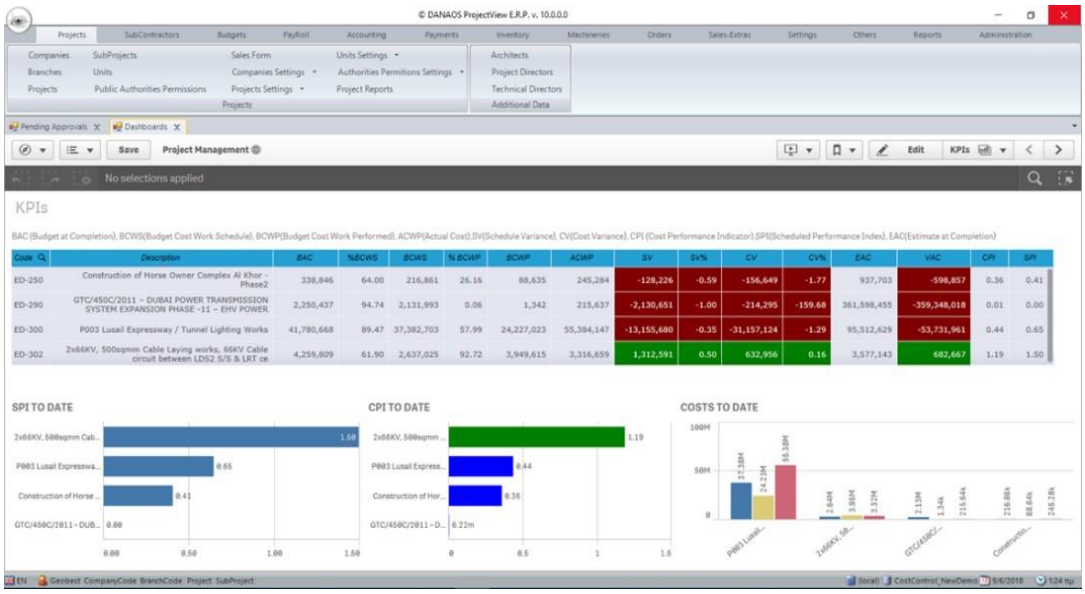

Figure 2. 1: The dashboard of DANAOS Constructions

Functional features

Functional features of DANAOS Constructions as below,

- Estimation management
- Cost Control
- Material Inventory
- Project viewing

Non-Functional features

Non-Functional features of DANAOS Constructions as below,

- User Friendliness
- Security
- Learnability

2.1.2 Sage construction management system

Sage Construction management system is another web-based software. This system also helped to the identify for the developed proposed system.

Following figure 2. 2 shows the dashboard of Sage construction management system

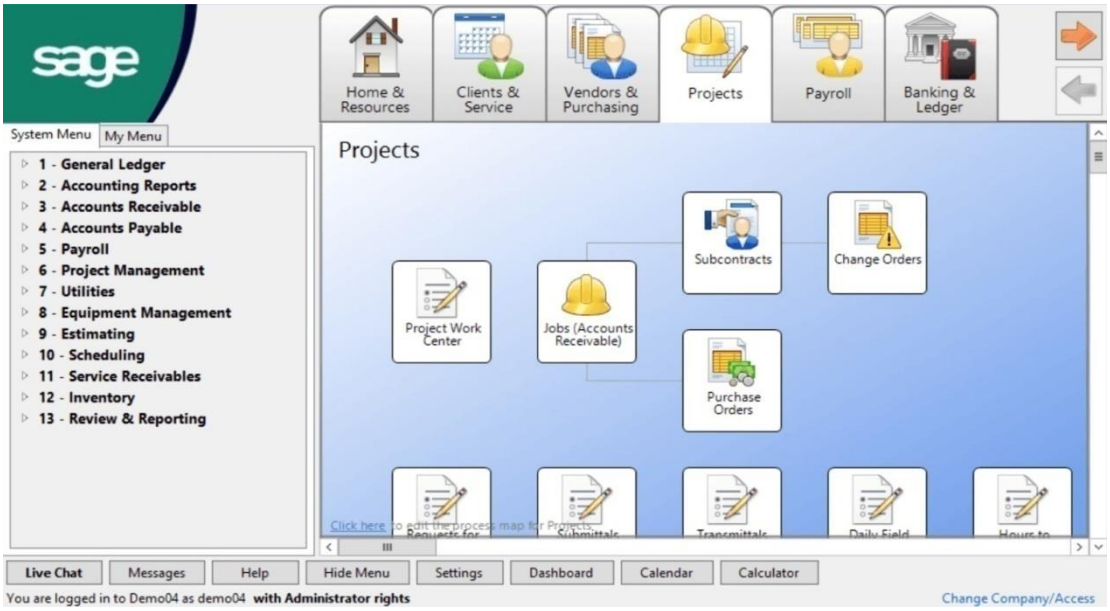

Figure 2. 2: The dashboard of Sage construction management system

Functional features

Functional features of Sage construction management system as below,

- Equipment management
- Estimation management
- Scheduling management
- Service Receivables
- Insightful Reports

Non-Functional features

Non-Functional features of Sege construction management as below,

- User Friendliness
- Simplicity

Above mentioned those similar systems, so many help to the clearly conduct to the proposed system. So, the UI design patterns, languages, front-end developing side were used the different ways. But most functional requirements are same.

Following table 2.1 shows the comparison between the above similar systems.

Features	DANAOS construction management system	Sage construction management system
Web-based system	yes	yes
Service Receivable	no	yes
Ability for the calculation to estimation	yes	yes
Manage the equipment inventory	no	yes
Employee scheduling	no	yes

Table 2. 1: the comparison between the above similar systems

2.2 Existing system

The Ceylon Engineering company currently carry out their process through the paperbased manual system. It is help for the customer, service reservation, estimation & employee sides. But this process very time consuming, lot of paper waste & lost some important documents. For the example, the organization, all the calculation of estimation doing by the calculators. All the details are recording in paper & store in the relevant file. The management need to get some reports of the estimation, it took get more time.

Following figure 2. 3 shows the top level usecase diagram for existing system.

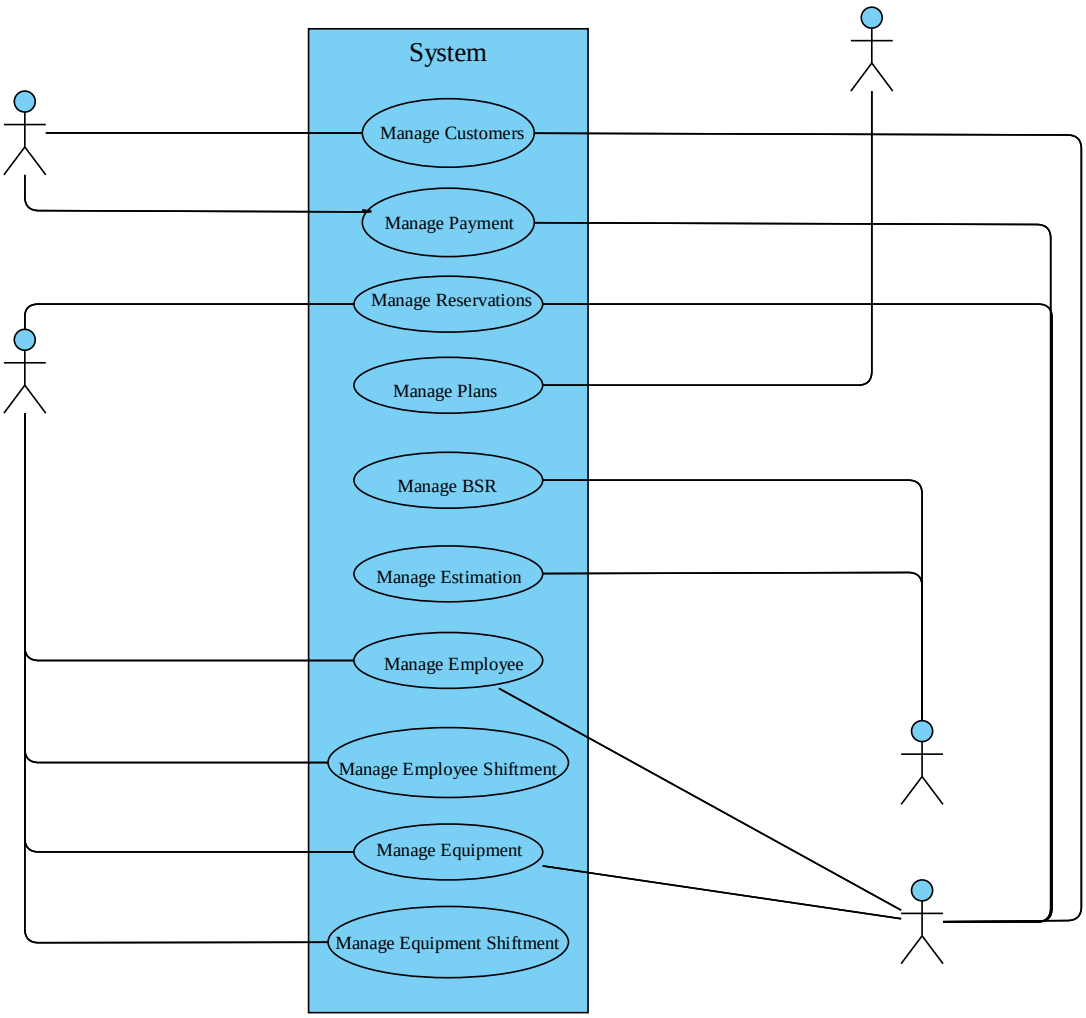

Figure 2. 3: The top level usecase diagram for existing system

2.3 Analysis of the requirements

Requirements analysis, also called requirements engineering, is the process of determining user expectations for a new or modified product. These features, called requirements, must be quantifiable, relevant and detailed. In software engineering, such requirements are often called functional specifications. Requirements analysis is an important aspect of project management. [2]

Requirements analysis involves frequent communication with system users to determine specific feature expectations, resolution of conflict or ambiguity in requirements as demanded by the various users or groups of users, avoidance of feature creep and documentation of all aspects of the project development process from start to finish. Energy should be directed towards ensuring that the final system or product conforms to client needs rather than attempting to mold user expectations to fit the requirements. [2]

Requirements analysis is a team effort that demands a combination of hardware, software and human factors engineering expertise as well as skills in dealing with people. [2]

2.3.1 Functional requirements

A Functional Requirement (FR) is a description of the service that the software must offer. It describes a software system or its component. A function is nothing but inputs to the software system, its behavior, and outputs. It can be a calculation, data manipulation, business process, user interaction, or any other specific functionality which defines what function a system is likely to perform. Functional Requirements in Software Engineering are also called Functional Specification. [3]

Functional requirements of the system as below,

- Customer Management
  Customer's main information are added the through the in this module.

- Service Reservation Management
  Reserve the service for the particular customer for in this module. And as well as record the plan and estimation details in each reservation.

- Service Plan Management
  The calculate square feet for each house subparts in this module. It will be help for the calculate the estimation for each reservation.

- BSR Management
  Maintaining the record of different building scheduling rate for each province and every year.

- Estimation Management
  Calculating the estimation for each reservation in this module.

- Employee Management
  Employee's main information are added through the in this module.

- Employee Shifting Management
  The employee can assign, remove or transfer to each reservation.

- Equipment Management
  Maintaining the recorded of all the equipment details in this module.

- Equipment Shifting Management
  The equipment can assign, remove or transfer to each reservation through the this module.

2.3.2 Non-functional requirements

Non-Functional Requirement (NFR) specifies the quality attribute of a software system. They judge the software system based on Responsiveness, Usability, Security, Portability and other non-functional standards that are critical to the success of the software system. [4]

Following the Non-functional requirements as below,

- User friendliness
  User interface of the system not be complicate. The user can easy to navigat and system should be implemented according user's point of view.

- Efficiency
  The system can operate the day-to-day business transaction efficiently without and delay.

- Security
  The system should prevent the unauthorized access and modification. It can be increase through privilege of the system. Privileges are granted for the only authorized users.

- Accuracy
  The system should allow the valid user input data. Hence the front-end and back-end validations are must be implement more accuracy.

- Reliability
  The system should operations of percentage completed correctly. Otherwise, the trustful information has to be used through the system.

2.4 Process model

There can be find out the different software development methodologies in the present. Waterfall model, Prototype model, Spiral model, Iterative and incremental model are example for that. Development models. Following evaluate these software development models as below,

2.4.1 Waterfall model

Waterfall model is introduced as a first process model in history. It is representing the linear sequential life cycle model. In waterfall model each phase should be completed the before attempt the next phase. And it will be helping the reduce the overlapping in the phases. These phases are called by the Analysis, Design, Implementation and Maintenance.

2.4.2 Prototype model

Prototype model is main or first version in final software. The prototype model is include the building, testing and reworking of model output. Otherwise, the prototype works like a version with various tests are involved and improvements are added.

2.4.3 Spiral model

The spiral model is the combination of the iterative development and systematic controlling of part of the waterfall model. Otherwise, the spiral model is a combination of iterative development process and sequential linear development. The spiral model has include the four phases. There are Identification, Design, Constructor or Build, Evaluation and risk.

Justification to select process model for proposed system

To select the process model for this proposed system firstly, should be determined the what is the scope of the project, what are the functional & non-functional requirements & identify the problem domain of the project. In this proposed system most suitable for the Iterative & Incremental process model. Because the requirements are most likely change and if the requirements are changed in while processing new requirements can be amplified without any issues. Therefore Iterative & incremental process model is the most appropriate for this project.

Iterative and incremental development (IID) is a model, that is developed multiple cycles of iteration. The project in the beginning with comparably small task(components) and

increments are made for each cycle of the iterations until agreeable product is reached. In the during this process developers must evaluate and testing those components at each stage and gathering information. It would be useful at the end of stages.

The Iterative and incremental development process model has four different phases. And each phases have different functions.

Following the phases as below,

- Inception – The main thing is in this phase identify the project scope & problem domain.

  Elaboration – In this phase mainly focus on architecture of the project. The MVC architecture was conducted for develop this project.
- Construction – The construction phase mainly considered as the project functionalities with design, programming & testing.
- Transition – In this final phase mainly focus on carry out the completed system to working environment.

Following figure 2.4 shows the iterative and incremental development process model

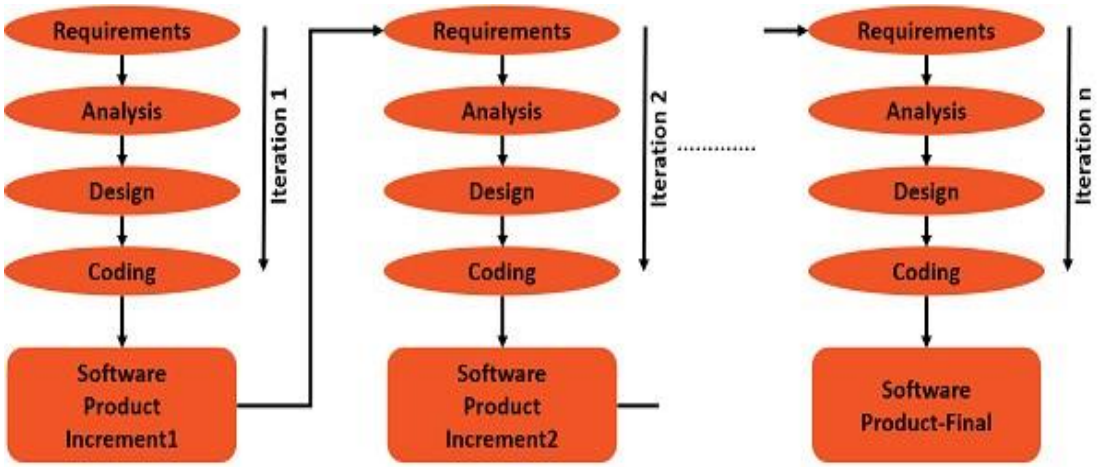

Figure 2. 4: The iterative and incremental development process model

Chapter 3 - Design

System design is a process of transform the user requirement into perfect form, which convenience the developer in coding and implementation phases. In system design phase deal with from the problem domain to the solution domain. Furthermore, in design phase should be consider the set of components, architecture, modules, interfaces for system to compatible with specified requirements.

3.1 Design strategies

Many different type of software development process models and design patterns can find out. But should be selected in those one of them for proposed system. Furthermore, standalone and web-based systems are mostly use. In the present as well as commonly use the mobile application and embedded systems.

Following mentioned about the differences between characteristics of them,

- Standalone system
  The Standalone system is runs on the locally machine. It doesn't need an internet connection and no need to browser to run like web-based system. Furthermore, the all data can be stored in locally on the device.

- Web-based system
  Web-based system can be accessed from anywhere, through the internet. The different type of platform can be used the web-based systems, because it only needs the web browser. Furthermore, it requires deployment on single set of server machine.

- Mobile application
  The mobile applications are most commonly use at the present. Because ease of use and it's portability. The initially mobile applications are provided the information services on the global network.

- Embedded system
  An embedded system is placed inside some kind of device to perform a very important set of functionalities.

Justification for the selected design strategy

Consider the above factors, the web-based strategy is the most appropriate suitable for this proposed system. Because any time can be access the system through the internet, it is more help for reducing the time wasting, as well as reduce the costs and more compatibility for the cross platforms. Typically, minimum require the web browser for the run the system.

3.2 Architectural design

The proposed system is developed through the extended MVC framework. The extended MVC is mostly use for the web application and it also refer the Object-Oriented design pattern. The extended MVC architectural pattern that divided application into three main logical components. Each component can perform the specific task in application.

- Model – The model component negotiates with the business logic related data. The main purpose of model component is the connection between with database and model class.
- View – The view component is related with UI logic in application. The users are interacting with this UI component.
- Controller – The controller creates the connection between model and view components for process all the business logic and requests.
- Repository – The repository is perform the query functionality of the model.

Advantages of MVC Design architecture

- The MVC follow the rapid and parallel development. The developer can work on the view while the other can work on the controller to build up the business logic of the system. Hence this way, can determine the faster applications developed using than other development patterns.
- Same component can be used or called for use the any interface.
- The MVC architecture compatible and integrate with JavaScript frame works.
  That mean MVC also supports an asynchronous technique.
- The MVC architecture provide the multiple view for model. Hence the code duplication is more reduce.

3.2.1 Use case Diagram

A use case diagram is a way to summarize details of a system and how to users are interactions with the system. It basically shows the graphic depiction and it through the determine the interactions among with different element in the system. Use case diagram will emphasize the events in system and different types of roles in a system.

Following information can be obtained from the use case diagram,

- Can be gather the requirements of the system.
- Can be recognized the internal & external factors about the system.
- Can be show the system-user interactions.

Recognize the type of users for the system is very critical. In this system, recognized type of users are following,

- Manager
- Sales Executive
- Supervisor
- Designer
- QS

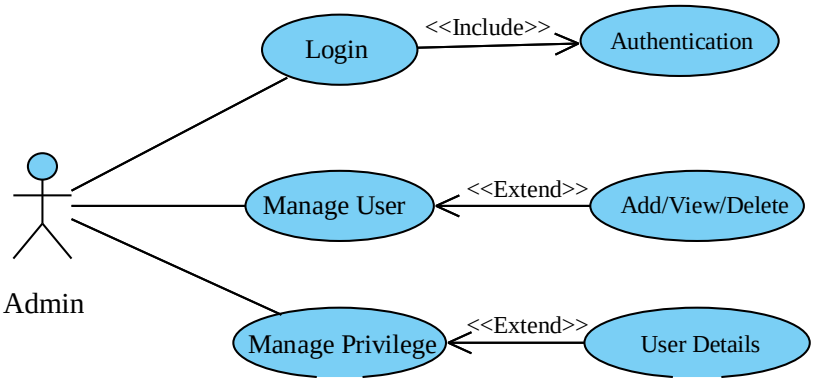

Figure 3. 1: Use case diagram for admin user

Following 3.1 table shows the use case narratives for user login

Use case	Login
Actor	All users (Manager Supervisor, Sales executive Designer, Qs).
Description	Authorized users can log into the system.
Preconditions	The user should register in the system and when login to the system user must enter a valid user name and password.
Flow of events	Enter a valid user name and password.
Post condition	If entered the valid user name and password the user will be showed the main window or will be redirected the login page and show the error message.

Table 3. 1: Use case narrative for user login to the system.

Following figure 3.2 shows the use case diagram for sales executive

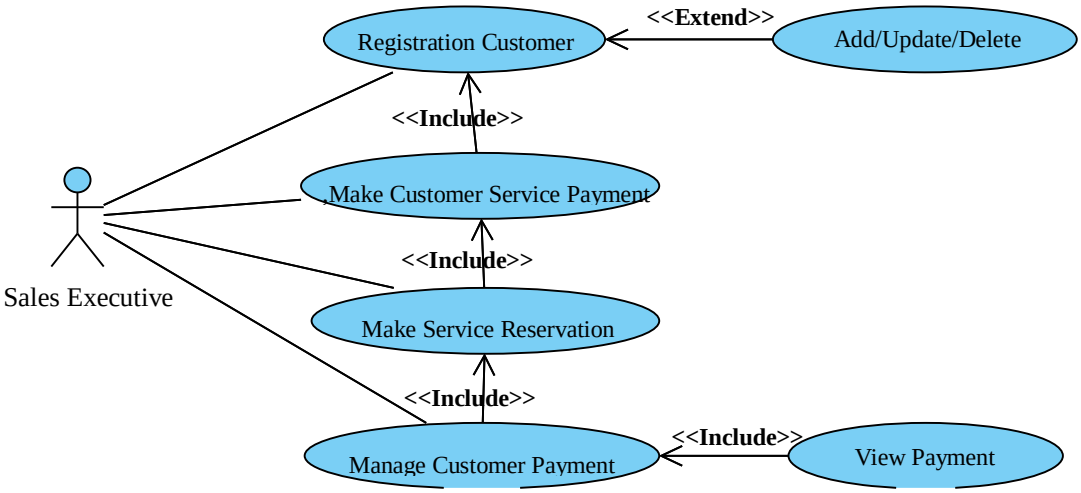

Figure 3. 2: Use case diagram for sales executive

Following 3.2 table shows the use case narratives for service reservation

Use case	Service Reservation
Actor	Sales executive
Description	Receive the service those who made the service payment
Preconditions	Only registered customers are allow to the make service reservation.
Flow of events	The sales executive login to the system then after select the registered customer and should be entered the valid reservation details.
Post condition	If the enter the valid details for reservation, it will be display the success message to inform that reservation is successfully added.

Table 3. 2: Use case narrative for service reservation

Following figure 3.3 shows the use case diagram for supervisor

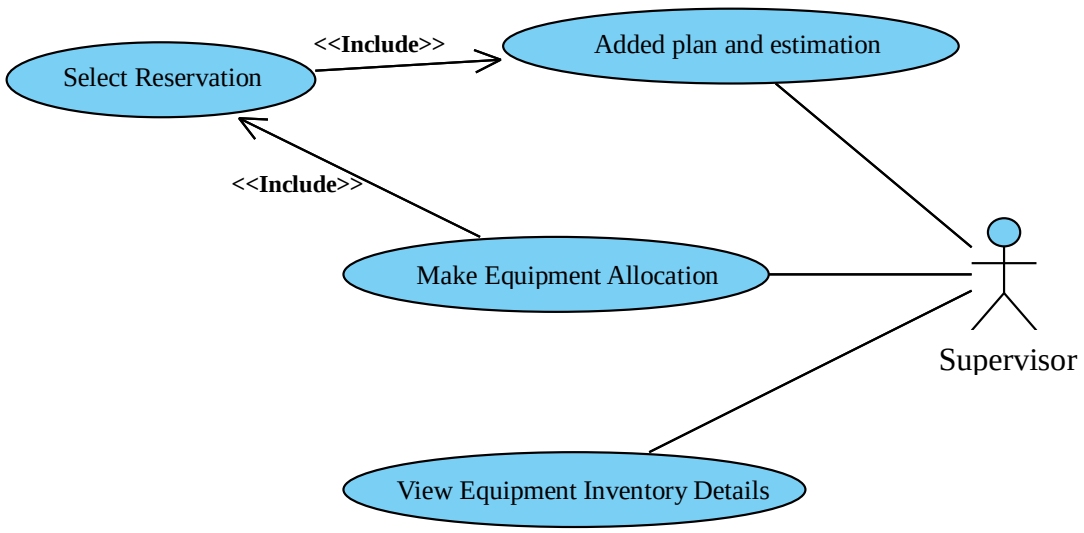

Figure 3. 3: Use case diagram for supervisor

Following 3.3 table shows the use case narratives for equipment allocation

Use case	Equipment allocation
Actor	Supervisor
Description	Equipment allocation can made only for received service reservation.
Precondition	The reservation must made for the service then after the plan and estimation should added to the reservation.
Flow of events	The supervisor chose the equipment allocation form then after select the reservation and assign, remove or transfer the equipment for reservation.
Post condition	If assign or remove the equipment from the reservation, then automatically updated the equipment inventory.

Table 3. 3: Use case narrative for equipment allocation

3.2.2 Activity Diagram

Activity diagram is representing the flow chart. In that flow chart describe the flow from one activity to another activity. And that activity can be informed the what are the operations of the system. Because it will show the message flow from on activity to another.

Following figure 3.4 shows the activity diagram for of login scenario

Following figure 3.4 shows the activity diagram for of login scenario

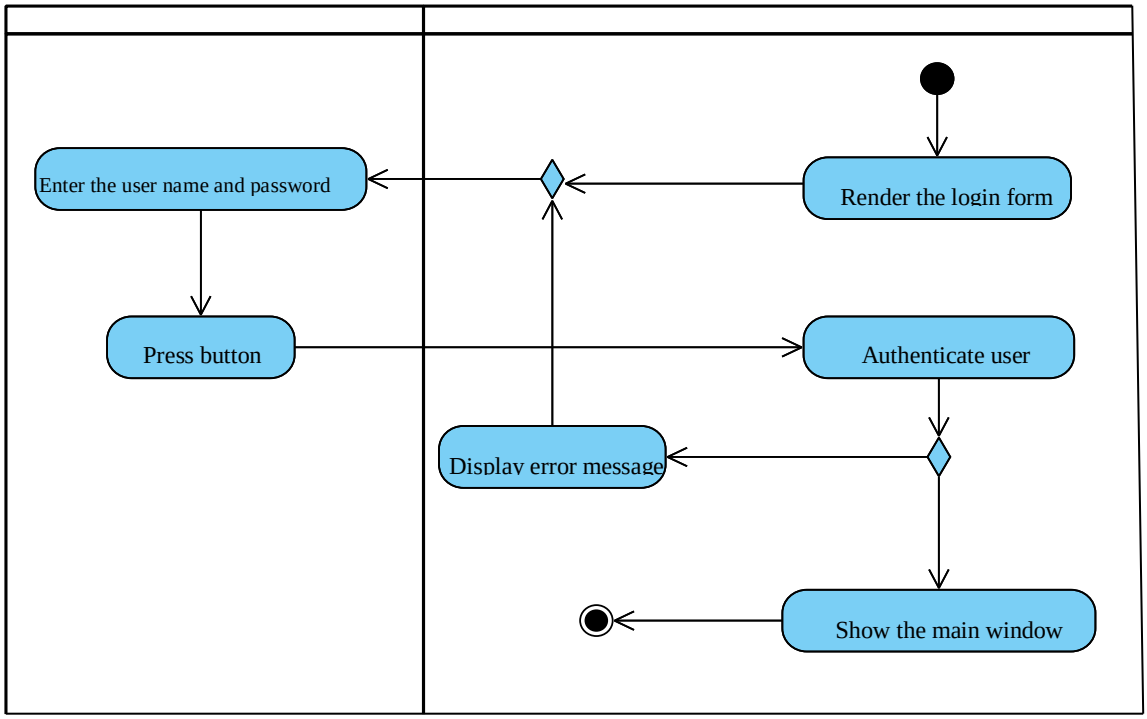

Figure 3. 4: Activity diagram for login scenario

Following figure 3.5 shows the activity display for service reservation scenario

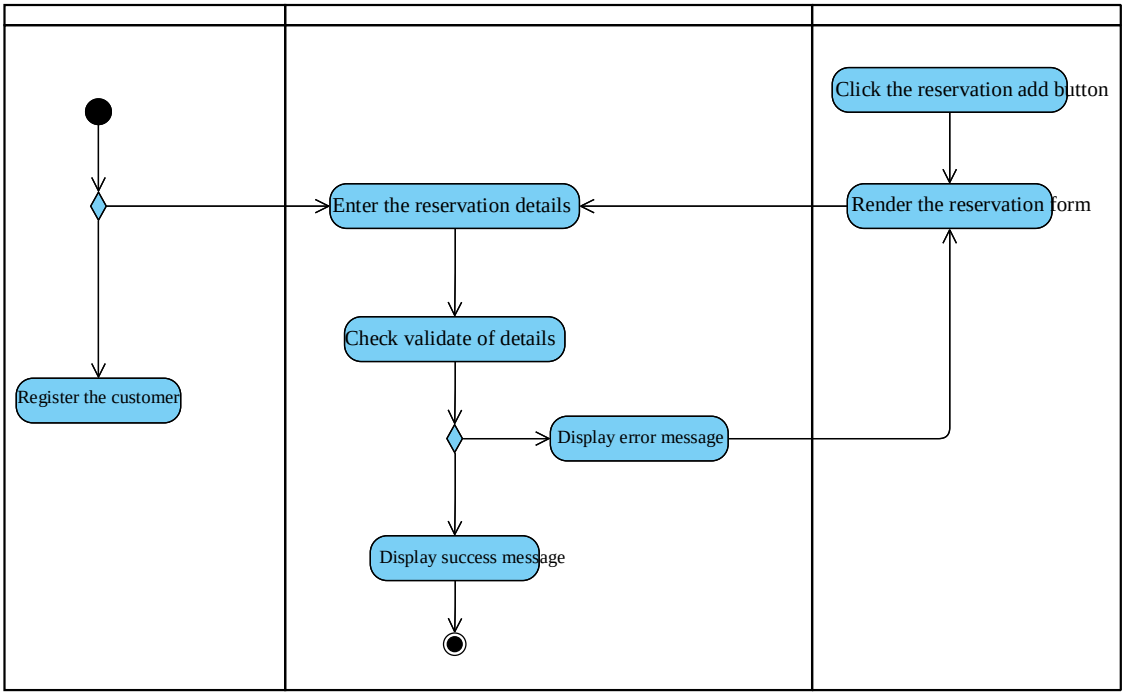

Figure 3. 5: Activity diagram for service reservation scenario

3.2.3 Sequence diagram

The sequence diagram is using the represent to interactions among objects in a single use case. It will show the different parts in a system interact with each part to carry out a function.

Following figure 3.6 shows the sequence diagram for service reservation scenario

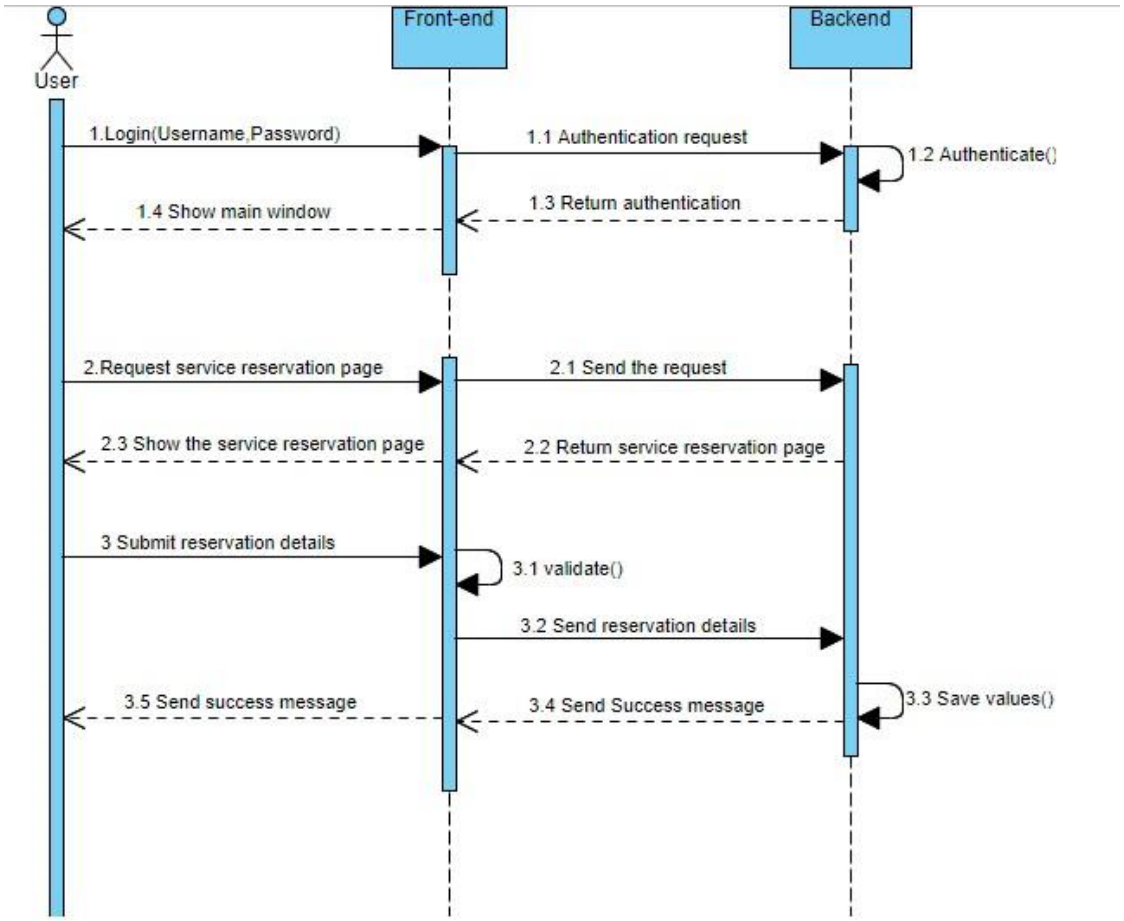

Figure 3.6: Sequence diagram for service reservation scenario

3.2.4 Class diagram

The class diagram is static diagram. It can be determining the static view of the system. And it also represents the attributes and operations of a class. The class diagram mostly used in the object-oriented modelling.

Following figure 3.7 shows the class diagram for entire system

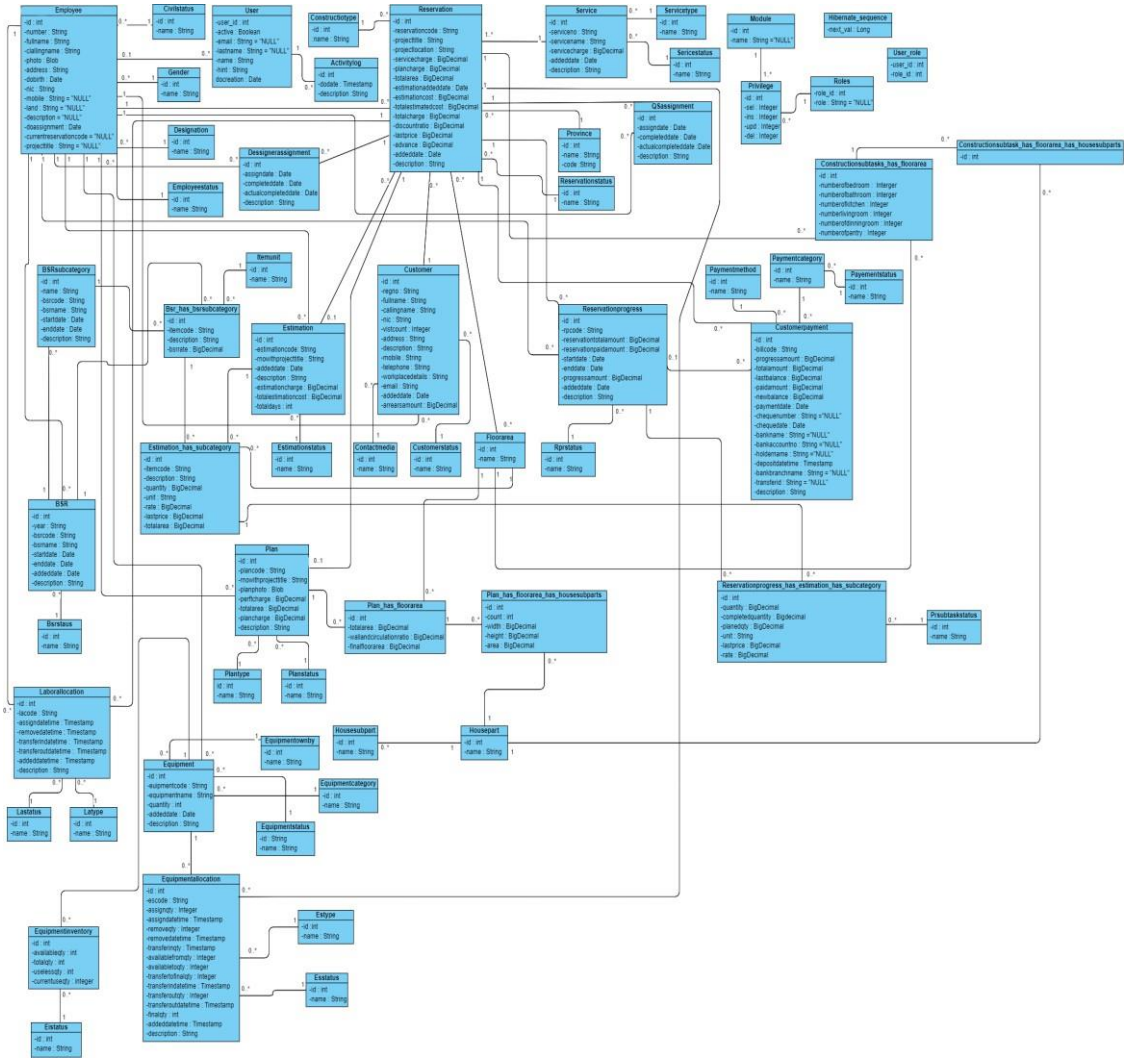

Figure 3. 7: Class diagram for entire system

3.3 Data modeling

Data modeling is the evaluate and identify the data objects and relationship to other objects. And also, data modeling use to analyze the requirements of data for the business process.

An entity relationship diagram is type of structural diagram for represent the database design. It describes the relationships of entity sets stored in a database and also shows the complete logical structure of a database.

Following figure 3.8 shows the entity relationship diagram for entire system

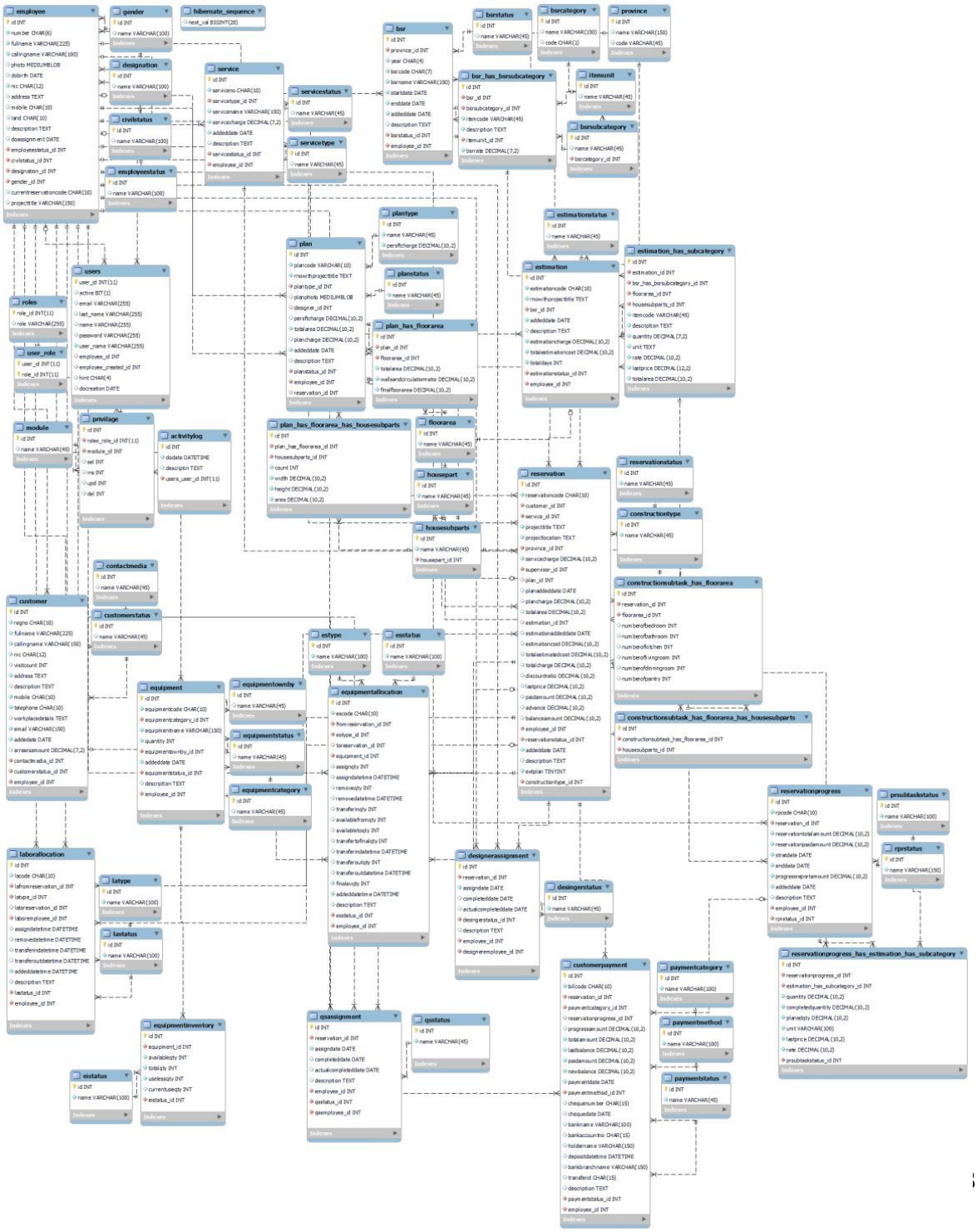

Figure 3. 8: Entity relationship diagram for entire system

3.4 User interface design

The main purpose of user interface design is effective interaction between a user and the system. The perfect user interface design provides the user-friendly experience and should be match with the user expectation. In the present all software programs have graphical user interface, or GUI. Because those software programs have graphical controls, which the user can choose the using a mouse or keyboard. Basically, GUI of software programs contain with a navigation bar, side bar, buttons and other controls.

Following figure 3.9 shows the user interface for login to the system. Firstly, the user interacts with the login page. Then it simply & clearly design structure has been use to better interact with user. The user should be enter the valid user name and password for login to the system. When user enters valid user name and password it will be automatically check the authentication of the user and allow to the log into to the system.

Figure 3. 9: User login interface

Following figure 3.10 shows the user interface of main window in system. It provides the collapsible side bar, rectangular tabs with links for access to system's main modules.

Hence user can be easy to navigate the system's main modules. Usually use the gray color & light blue color for design some UI components. Because it represents inherence of Ceylon

Engineering company. It will emphasis the basic mood, tone, concept & connotation in Ceylon engineering company. And also included the information about active reservations in here. Because it will help to know in advance what are the active reservations when the authorize user log in.

Figure 3. 10: Main window user interface

Following figure 3.11 shows the interface which will using for manage to service plan details. In service plan interface also has been use the clear & simply design structure. There are two inner forms in service plan interface for gathering the main details of house plan. Such as total area, sub total area. There are some disabled fields for the autogenerated values when calculate the total area and sub total area. And active and disabled buttons for certain conditions.

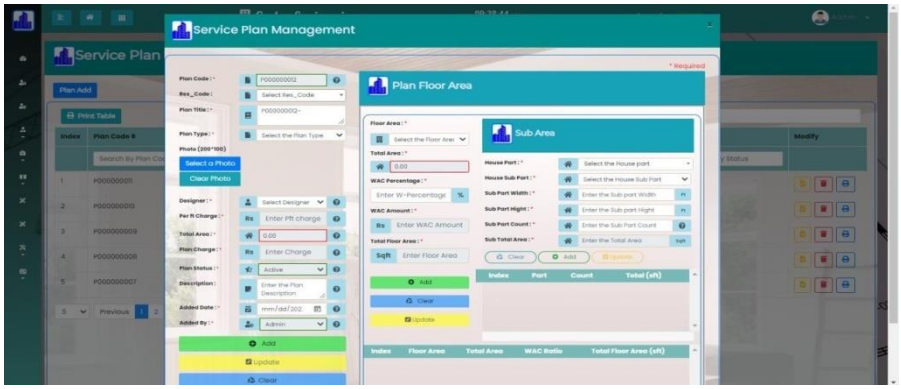

Figure 3. 11: Service plan user interface

Chapter 4 – Evaluation

The testing process is emphasis the evaluating of system performance. Otherwise, it describes the intend to find that whether it achieve the specific requirement or not. And also testing is identify the drawback, errors, misfits of actual requirements. In software testing verification and validation portions are very important things. Because these things are should be satisfied the specifications and fulfil the main objectives. Verification is followed the software should eligible in given development phase with accomplish the conditions at the beginning of that phase. Through Validation can be determine that product design satisfy or accomplish the intended use or the software reach the user requirements.

4.1 Description of testing approach

In system testing there has different type of test systems. There are unit testing, integrated testing, system testing etc. Testing is continuing while development progress going on. Therefor test cases and test data are used for examine.

4.1.1 Unit testing

The unit testing is a technique and it should be testing the individual units of source codes, set of one or more project modules together with an associated control data, usage progress and operation progress are tested to achieve if they are suitable for use. It can detect failures in algorithms and check for logical improvement in coding. And it more helps increase the efficiency for each unit.

4.1.2 Integration testing

In integration testing the group of components are compounded and tested, as a group. The integration testing is execute after unit testing done and before the system validation phase. Integration test plan execute as its test units are pre-tested by unit testing.

4.1.3 System testing

The system testing is implemented the whole system in the scope of functional requirement specification and/or system requirement specification. System testing should be satisfying

the client expectations. It also considered to test up to and beyond the limitations defined in the software/hardware requirement specifications. It is carried out the after integration testing by testers.

This is a black box type testing. The software is analyzed with the help of requirement documents and mainly based on user perspective. As a rule, the system testing acquires as its input, all of the “integrated” software units that have confirmed integration testing and also the system itself compounded with any appropriate hardware systems.

4.2 Test plan

The test plan is main portion of develop the system. Test plan should govern the functionality of the overall system. The good test planning can be determine the system’s errors and also can find the what parts need to be improved in the system. The system tests will aim on the behavior of the proposed system. Generally, the system test will test the integrated system and verify that fulfil requirements pre-defined in the requirements document. Security tests will resolve how secure the proposed system. Documentation tests will be accomplished to evaluate the accuracy and relevant of the user documentation. Through those tests will accomplish that no features are missing, and the all the contents can be clearly recognize.

4.3 Test cases

The clearly planned test case can be checking the relevant system component functionality. Before the check all system functionality, there should have properly planned test cases for every function. There are some test cases designed for main system modules as bellow,

4.3.1 Test case for login

Test No	**Test Description**	**Expected Result**	**Pass/Fail**
1	Click log in button without filling username and password.	Focus the username field and appear the “Please fill out this field” validation message.	Pass

2	Click log in button filling with incorrect username and incorrect password.	Give the error message "User Name or Password invalid, please verify"	Pass
3	Click log in button filling with correct username and password.	Appear the main window without display any errors.	Pass

Table 5. 1: Test case and test results for login

4.3.2 Test case for customer module

Test No	Test Description	Expected Result	Pass/Fail
1	Auto Generated Customer Id	Increment the last customer Id by 1. Should not be duplicated. Field border should be green.	pass
2	Customer "calling name" with no spaces. First letter is wanted to be capital.	Text field border color changed to green.	pass
3	Customer "full name" with can contain space. Contain any digit.	Text field border color changed to green.	Pass
4	Customer "NIC" length is 12 or length is 10 End with V.	Text field border color changed to green.	Pass

5	Customer “mobile” number start with 0 the length is 10.	Text field border color changed to green.	Pass
6	Select the contact media from the combo box.	All the contact medias are loaded into the combo box.	Pass
7	Customer “email” include with alphabetic characters, numbers, @	Text field border color changed to green	Pass
8	Assign date auto generate.	Current date appear in date field and field border should be green color.	Pass
9	Added by field the auto generate.	Added by field auto generate in when user is logged to the system	Pass
10	Customer status is auto generated.	Customer status appear in as “Active”. Customer status combo box border should be green color.	Pass

Table 5. 2: Test case and test results for customer

4.4 User acceptance testing

After the conclusion of implementing the system was testing performed by the end user or client environment. The main focus on user acceptance testing is validate end to end business flow. The client was recognizing main functionality of the proposed system. The briefly explaining about the system and client could be realize that how to system process flow continue. Generally, the acceptance testing was initiate by providing actual database and entering valid data to the system. And it was helped for client evaluate the overall system's process. The client was evaluate the performance and short comes and user friendliness of the system. Interview were accomplish the real users, system functionalities and user friendliness of the system. Over all observe of user response was complete as the final stage. Following figure show the several feedbacks given to the system.

Name: Mr. Prasanth Perera			Designation: Chairman		
Questions	**Very Bad**	**Bad**	**Average**	**Good**	**Very Good**
Your level of computer and internet literacy				✓	
User friendliness of the user interface				✓	
The system demonstration performed				✓	
Level of fulfilment of the stated objectives					✓
Your level of satisfaction about this system					✓
How do you categorize the functionality and the usability of the following features given by the system?					
Login management					✓
Customer management					✓
Reservation management				✓	
Service plan management				✓	
Estimation management					✓
Service management					✓
Equipment management				✓	
Equipment shifting management			✓		
Employee management				✓	
Employee shifting management			✓		
Customer payment management				✓	

5. 1: The several feedbacks given to the system

Chapter 5 – Conclusion

In this chapter will evaluate the deviations of essential project requirements and what are the future improvements of the project.

5.1 Critical assessment of the project

The proposed system was accomplished to simplify the reservation, equipment and employee allocation activities for construction of Ceylon engineering. Basically, the business process was initiate with register a customer for certain pre-defined services. In this customer module hold and maintaining the essential data of the customer. Then after make the reservation for these services. In order to reservation module and service plan module are helped continue the main process of the system. Otherwise, the reservation module manages the essential details such as house subtask of service as well as through the service module can be measure the such as total feet of house subparts, total area of floor areas. And it more contributes to making the estimation for each plan.

The equipment and employee allocation modules were performed the major process in system. Because assignment, removal and transfer of equipment and employees every operation is monitored by this allocation modules. And it also maintains equipment inventory through this system. It will be helped for the to know about equipment current usage. The management staff of the company have motivated and working process accuracy and efficiently after introducing the system as well as the company enhanced their business (making reservations) as their activities have been execute the smartly and efficiently.

5.2 Problem encountered and solutions

In the requirement gathering phase, it was difficult to gather the information from the client. Because new requirements regularly added during the development process. Therefor changing the requirements and as well as in implementation phase also changed. In the implementation phase there was difficult to use some business scenario to be practically. Because some house parts and house subparts details were more complicated.

And sometimes those details were difficult to handle through some modules, such as service plan module and estimation module.

Another problem occurred. That is to make the system a completely systematic process. Because of completely manual system tasks were difficult to convert to an automated system.

5.3 Future improvements

It was difficult to determine if the system was 100% complete with client expectations for develop. But major functions and major requirements should be satisfied with the client expectation. And it was enhance the standard of the system and sustainably may be match with actual business scenario. Following as mention the future improvements of the system,

- Providing the online payment facility to user. By this, the user can make their transaction through the system using a payment gateway.
- The system can implement and generate more reports and charts. And it helps the decision making in for management of company.

5.4 Lesson learnt

As a BIT student this project program was more helped to enhance about the knowledge of different technologies as well as got the lot of experience. The project domain was significant large. Therefore, it more contributes to determine the different development methods for business process and can be determine the how to resolve the problems in business scope during developed the system.

The various type of communication techniques learned while doing questionnaires and interviews through the company management.

The spring boot framework, Java language, MySQL, IntelliJ IDEA were more helped the enhance technical knowledge again and again. Through the analyzed manual system process, business analysis skills were developed the perfectly. As well as the design skills was enhanced through the sort of diagrams. Such as Use case diagram, Activity diagram, Sequence diagram, Entity relationship diagram. Another main things, the coding skills, codding error resolving problem resolving, logical thinking patterns were enhanced through developed the system.

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